

Kyle Wong

Recent Master's graduate from Sidney Sussex College, University of Cambridge

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A highly motivated recent Master's graduate from the University of Cambridge with a strong background in astrophysics, mathematics, and computational science. Experienced in cosmological research, gravitational wave data analysis, and radio astronomy. Seeking a challenging research or data science role to further develop expertise in scientific computing and data analysis.

EDUCATION

Sidney Sussex College, University of Cambridge, Cambridge, England, United Kingdom 2024 – 2025

MASt/Part III in Astrophysics

Graduated with Good Honours equivalent

Victoria College, University of Toronto, Toronto, Ontario, Canada 2020 – 2024

HBSc in Physics & Mathematics

Graduated with High Distinction

German Swiss International School, Hong Kong SAR, China 2010 – 2020

International Baccalaureate (IB) Diploma in 6 subjects

International Advanced Subsidiary (AS) level in 1 subject

International General Certificate of Secondary Education (IGCSE) in 12 subjects

RESEARCH EXPERIENCE

Master of Advanced Study Research Project Cambridge, England, United Kingdom

Postgraduate Student

October 2024 – September 2025

Supervised by Prof. Anastasia Fialkov at the Institute of Astronomy, University of Cambridge

- Research topic: implementation and statistical testing of variable initial conditions (cosmologies) in 21cmSPACE code package for simulating the distribution of hydrogen gas clouds in the early universe, making possible efficient forecasting of future radio astronomy experiments, most notably for the Square Kilometre Array (SKA)

Canadian Institute for Theoretical Astrophysics

Summer Undergraduate Research Fellowship

Toronto, Ontario, Canada

Undergraduate Researcher

May 2023 – December 2023

Supervised by Dr. Philippe Landry at the Canadian Institute for Theoretical Astrophysics

- Research topic: probing neutron star tidal deformability from gravitational wave signals using Markov chain Monte Carlo (MCMC) parameter estimation in many dimensions, and incorporating new models for neutron star equation of state correlations in the analysis pipeline of Laser Interferometer Gravitational-Wave Observatory (LIGO) Scientific Collaboration gravitational wave data

McGill Space Institute Summer Undergraduate Research Award

Montreal, Quebec, Canada

Undergraduate Researcher

May 2022 – April 2023

Supervised by Prof. Adrian Liu at the Trottier Space Institute (formerly McGill Space Institute) of McGill University

- Research topic: incorporating statistical priors into the power spectrum data estimator used in the data pipeline, for analysis of radio astronomy data from the Hydrogen Epoch of Reionization Array (HERA) collaboration's cosmic dawn experiment, using the Python programming language

HONOURS, AWARDS AND SCHOLARSHIPS

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|---|------------------|
| • Dean's List Scholar in the Faculty of Arts & Science | 2022, 2023, 2024 |
| • Canadian Institute for Theoretical Astrophysics
Summer Undergraduate Research Fellowship (CITA SURF) | 9,500 CAD, 2023 |
| • Birkenshaw Family Scholarship II | 1,000 CAD, 2023 |
| • McGill Space Institute (now Trottier Space Institute)
Summer Undergraduate Research Award (MSI SURA) | 7,000 CAD, 2022 |
| • David and Louise Fraser Scholarship | 2,500 CAD, 2022 |
| • University of Toronto Scholar | 1,500 CAD, 2022 |
| • Birkenshaw Family Scholarship | 1,000 CAD, 2022 |

SKILLS

- **Technical:** Scientific computing, machine learning, statistical inference, simulation, time-series analysis, data visualization, high-performance computing.
- **AI Tools:** Claude Code, Google Gemini, Gemini CLI, ChatGPT, and Google Antigravity for automated software engineering, academic and scientific research, and coding.

PRESENTATIONS

- | | |
|--|------|
| • Interim progress presentation, at the Cambridge Cosmic Dawn Group | 2024 |
| • <i>Estimating Neutron Star Tidal Deformability</i> , at the CITA Undergraduate Research Showcase | 2023 |
| • PHY478 Physics Project final presentation | 2023 |
| • Two presentations given at the CITA Compact Objects Group, as part of PHY478 Physics Project | 2023 |

FEATURED PERSONAL PROJECTS

mercury_strategies [Private Repository] (work in progress)

- Engineered a platform-agnostic trading system for weather-derivative prediction markets (e.g., Polymarket) using an asynchronous event-bus architecture in Python. Implemented a watcher-based service for real-time polling of weather providers and high-frequency market state through WebSockets and local orderbook monitoring; deployed dockerized on AWS for 24/7 automated operation.

quant_core (https://github.com/kyleyhw/quant_core)

- Architected a market-agnostic algorithmic trading framework using a Market Adapter pattern to decouple core logic from broker-specific APIs. Implemented a quantitative pipeline for time-series forecasting and regime detection using Hidden Markov Models (HMM) and XGBoost.

sound_simulation (https://github.com/kyleyhw/sound_simulation)

- A numerical simulation of physical waves in N-dimensions, implemented in Python. The simulation allows for the creation of physical boundaries and obstacles, and models their interaction with the waves. Future work aims to use this simulation to generate training data for machine learning models.

SELECTED COURSES

- Gravitational Waves & Numerical Relativity (Cambridge, 2025)
- Astrostatistics (Cambridge, 2025)
- Cosmology (Cambridge, 2024)
- General Relativity (Cambridge, 2024)
- General Relativity (Toronto, 2024)
- Relativity Theory II (Toronto, 2024)
- Relativity Theory I (Toronto, 2023)
- Physics Project (Toronto, 2023)
- Computational Physics (Toronto, 2023)
- Computational Astrophysics (Toronto, 2023)
- Advanced Classical Mechanics (Toronto, 2023)
- Geometry of Curves and Surfaces (Toronto, 2023)
- Partial Differential Equations (Toronto, 2023)

RESEARCH INTERESTS

- Computational data analysis
- Simulation and numerical methods
- Machine learning, artificial intelligence
- Astrophysics (including gravitation, cosmology, black holes & compact objects, dark matter, dark energy)

RELEVANT EXPERIENCE

- **Teaching:** Private Tutor for physics & mathematics (IGCSE, A-levels, IB syllabi). 2020 – 2022, 2024 – 2025
- **Professional Development:** Intensive astrophysics bootcamp at the University of Pennsylvania. 2022
- **Volunteering:**
 - Involved in public astronomical observing nights at the Institute of Astronomy.
 - Involved in Beaver Scouts outreach visits at the Institute of Astronomy.
 - Volunteered for RiseWise HK, serving as an assistant soccer coach for children with special educational needs.
 - Travelled to Chiang Mai, Thailand to physically contribute to the construction of a school.
 - Undertook training and assisted with a public astronomical observing night at McGill University.

LANGUAGES

- English (fluent, language of education)
- Cantonese (fluent, mother tongue)
- Mandarin (formally learned 10 years)
- German (formally learned 10 years)

RECENT MEMBERSHIPS

- Sidney Sussex College Football Team, Premier Division
- Cambridge University Mountaineering Club
- Cambridge University Table Tennis Club
- Cambridge University Hong Kong Postgraduate Student Association
- Cambridge University Chinese Society

PERSONAL INTERESTS

- Rock climbing (9+ years experience both indoors and outdoors, with awards won at amateur, inter-high school and inter-university competitions. Also led a rock climbing extra-curricular activity during high school, including captaining the competition team)
- Soccer (7+ years league participation)
- Scuba diving (PADI open water certified)
- Travelling